



Example 1

A car main gear shaft is made using a 18NiCrMo4 steel. Some operations are done by using a lathe, using a between centers configuration.

The first operation brings the diameter of the part from 48 mm to 40 mm. The stock is 180 mm long. This operation is divided in two steps: roughing and finishing.

Operation data: $k_{c0,4} = 2.800$ MPa, coefficient $x = 0,29$, entering angle of the main cutting edge $\kappa_r = 35^\circ$, entering angle of the secondary cutting edge $\kappa_r' = 55^\circ$

- **Roughing.** The tool used has a tip radius $r_\epsilon = 1,2$ mm:

Feed	Depth of cut	Cutting speed
$f = 0,5$ mm/rev	$a_p = 3$ mm	$v_c = 170$ m/min

- **Finishing.** The tool used has a tip radius $r_\epsilon = 0,4$ mm :

Feed	Depth of cut	Cutting speed
$f = 0,2$ mm/rev	$a_p = 1$ mm	$v_c = 325$ m/min



Example 1

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The lathe has a nominal power of 25 kW and a 90% efficiency; the maximum number of rpm is equal to 5000 rpm.

It is required to:

- a) Check if the machine specifications are compatible with the execution of the required machining operations.
- b) Determine the theoretical roughness after the finishing step.
- c) Determine the machining time for the finishing step, considering an overtravel of 2 mm before and after the operation.



Example 2

A cylindrical component is machined using a lathe.

Process parameters:

- Tip radius $r = 0,4 \text{ mm}$
- Roughness $R_a = 3,2 \text{ } \mu\text{m}$
- Depth of cut $a_p = 2 \text{ mm}$
- Specific cutting force $k_{c 0,4} = 2100 \text{ MPa}$
- Constant $x = 0,29$
- Entering angles: $\kappa_r = \kappa_r' = 45^\circ$
- Taylor's Law parameters: $C = 250, n = 0,122$

Machine parameters:

- Nominal power: 6 kW
- Efficiency: 80%
- T_t : 120 s

a) Determine the cutting speed that maximizes the throughput.

b) Determine the maximum feed, considering all constraints.



Example 3

A component is machined using a lathe from a steel bar, using a between centers configuration. In the following table, some machining parameters are reported.

Product data	Process Parameters	Process times	Costs
$L = 200 \text{ mm}$	$f = 0,5 \text{ mm/rev}$ $\kappa = 45^\circ$	$T_t = 60 \text{ s}$	$C_0 = 100 \text{ €/h}$
$D_0 = 40 \text{ mm}$	$n = 0,122$ $C = 230$	$T_h = 30 \text{ s}$	$C_t = 5 \text{ €}$
$D_f = 36 \text{ mm}$	$k_{c0,4} = 2600 \text{ MPa}$ $x = 0,29$		



Example 3

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Determine:

- a) Cutting power with a cutting speed of $v_c = 130$ m/min;
- b) Cutting speed such that it is possible to make exactly 10 parts;
- c) Check production costs of points a) and b);
- d) Cutting speed that minimizes the production cost, and, therefore, the optimal cost.



Example 4

A shop floor produces steel cylindrical shafts. An external cylindrical turning is done using a between centers configuration. It brings the diameter of the part from 12 mm to 10 mm. The part is 70 mm long.

Machining parameters:

- *Roughing:*
 - Depth of cut $a_p = 0,8 \text{ mm}$;
 - Feed $f = 0,2 \text{ mm/rev}$;
 - Cutting speed $v_c = 150 \text{ m/min}$;
 - Extra times = 2 s;
 - Tool life = 10 min.
 - Tool change time = 3 min.
 - Load/unload time = 10 s.
- *Finishing:*
 - Total time = 10 s;



Example 4

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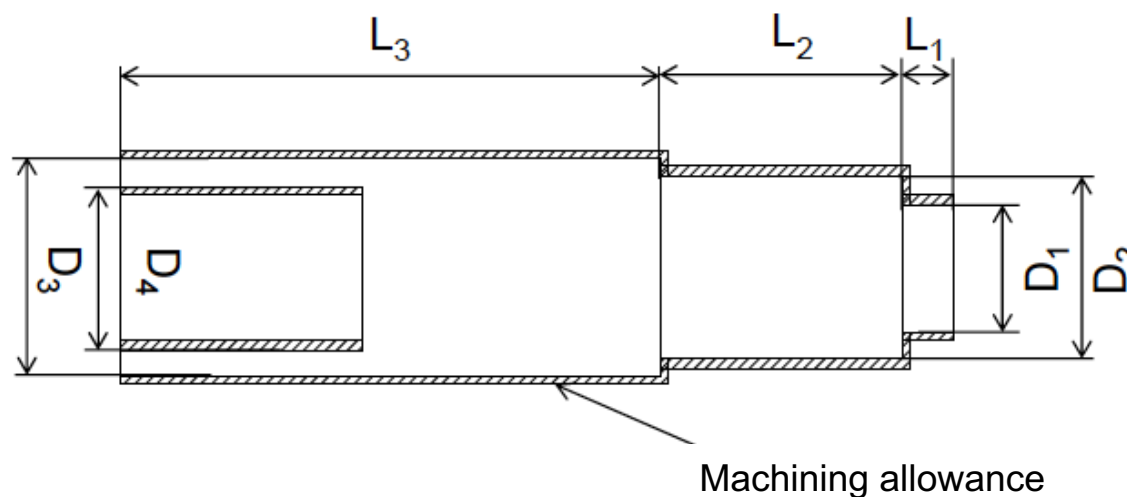
Determine:

- a) Contact time for the roughing operation;
- b) Roughing tool life, using a cutting speed of 130 m/min , knowing that if the cutting speed is reduced at 125 m/min , the corresponding tool duration is equal to 25 min ;
- c) Number of shafts produced in one shift of 8 hours, using a cutting speed of 150 m/min (assume infinite tool life for finishing and cutoff operations). The setup time is equal to 15 min, at the beginning of each shift.



Example 5

The figure reports the components to be machined starting from a steel stock. The dimensions are reported in the table. All sizes are referred to the finished part. Moreover, taking into account the interaction between cutting tool and material, the x and $k_{c0,4}$ coefficients are equal to 0,29 and 2100 MPa respectively.



Dimension	Value [mm]
D1	20
D2	25
D3	30
D4	24
L1	30
L2	70
L3	700

The machining allowance is equal to 1 mm and will be removed with a single finishing step. The average roughness R_a should be lower or equal than $1,5 \mu\text{m}$. The tool used for the finishing has a tip radius equal to 0,8 mm and an entering angle of the primary cutting edge equal to 95° and a secondary one equal to 45° .



Example 5

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The fixturing system consists in the usage of a self-centering three-jaws chuck, from the inside of the hole on the left. Using this fixturing, the finishing operation will be done on the external surfaces with a constant rpm, equal to 3000 rpm.

It is required to:

- a) Determine the contact time for the external finishing operation.
- b) Check the feasibility of the fixturing system for that operation.
- c) Considering two experimental cutting speed, namely v_{c1} and v_{c2} , and two experimental tool life values, namely T_1 and T_2 , determine the external finishing operation duration if the adopted cutting speed is set to the optimal value with respect to production time.



Example 5

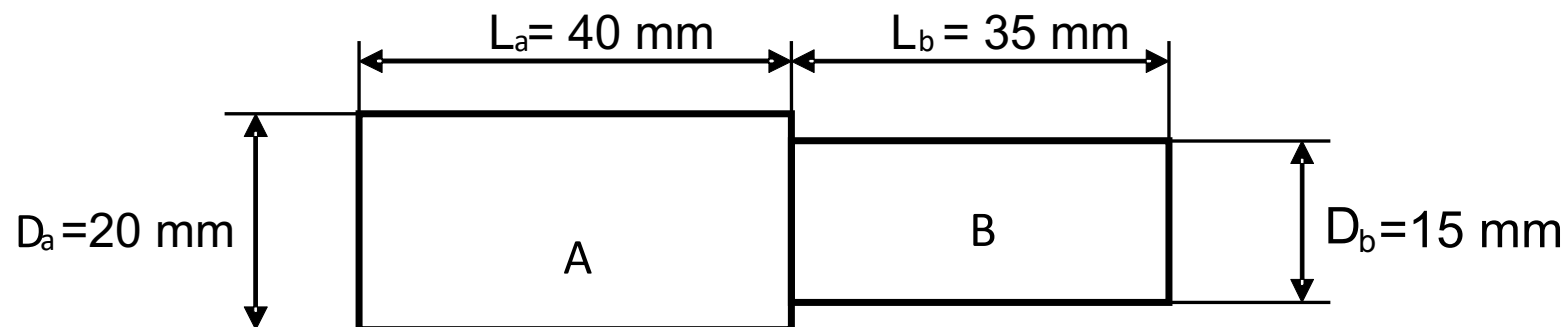
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Friction coefficient	0,45
Number of jaws	3
Jaw-workpiece contact area	400 mm ²
Maximum contact pressure	100 MPa
V_{c1}	280 m/min
V_{c2}	310 m/min
T_1	19 min
T_2	12 min
T_t	60 s



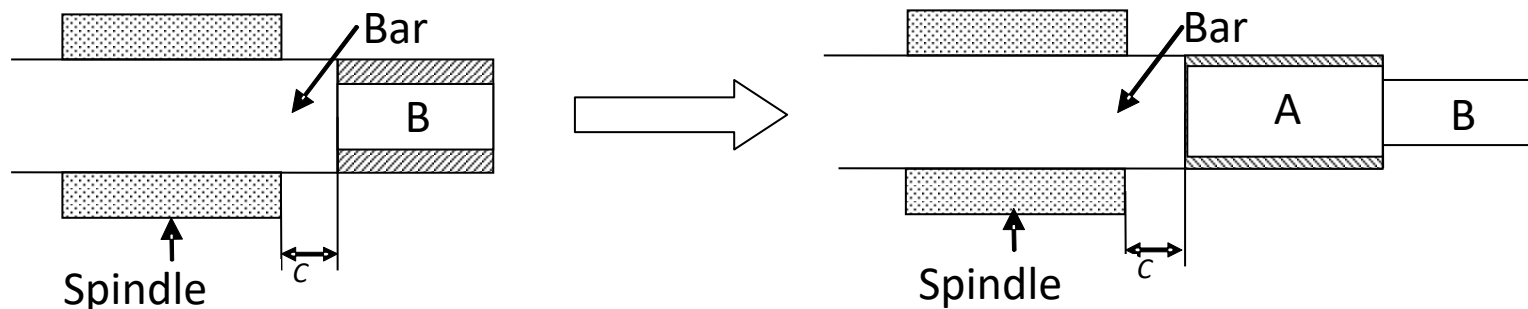
Example 6

The following component should be manufactured using a lathe, starting from a metal bar of diameter 25 mm. The bar is mounted using a lathe chuck.



The part can be machined by using four different strategies:

- **Strategy 1.** The length B is machined in one step. Then, the tool is disengaged from the part and the bar is repositioned as shown on the right side. The length A is machined in one step. The feed value is equal to 0,2 mm/rev, while the repositioning time is equal to 2 s.

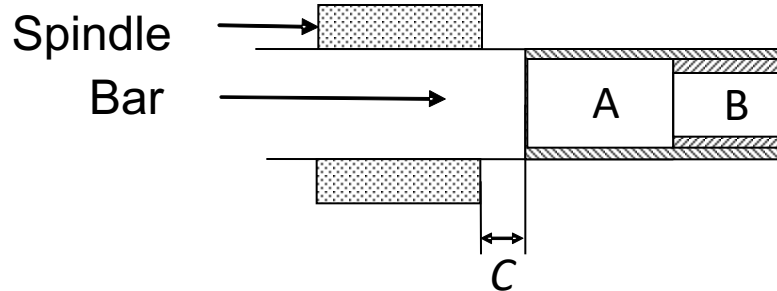




Example 6

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- **Strategy 2.** Same as Strategy 1, with 0,1 mm/rev as feed.
- **Strategy 3.** The bar is turned along the entire length at a diameter D_A , then, the tool is repositioned and the length B is machined to a diameter D_B . Feed is equal to 0,2 mm/turn. The repositioning time is equal to 2s.



- **Strategy 4.** Same as Strategy 3, with feed 0,1 mm/turn.

It required to:

- Determine the feasibility of each strategy.
- For feasible strategies, determine the maximum bending.
- State which one, among the feasible strategies, has the highest throughput (assuming the same values for fixed times).



Example 6

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Maximum RPM	6000 rpm
Efficiency	0,95
Kronenberg coefficient x	0,29
$k_{c0,4}$	2200 MPa
distance “c”	5 mm
Young’s module E	206000 MPa
Cutting speed	260 m/min
Entering angle of main cutting edge	90°
Overtravel	2 mm
Nominal lathe power	7,6 kW